

DEPARTMENT OF COMPUTER SCIENCE

Master of Computer Science

Semester – 3

SUBJECT: ENTERPRISE JAVA TECHNOLOGIES

Date: 10-08-2026

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ASSIGNMENT – 1

JDBC-ODBC AND SERVLET

Part A – JDBC / JDBC-ODBC

1. Explain JDBC. Discuss its purpose and importance in Java database application development.
2. Explain the architecture of JDBC with a neat and well-labelled diagram.
3. Explain the different components of JDBC architecture and describe their roles.
4. What is JDBC-ODBC connectivity? Explain the concept and working of the JDBC-ODBC Bridge.
5. Explain the different types of JDBC drivers. Compare Type 1, Type 2, Type 3, and Type 4 drivers.
6. Explain the steps involved in establishing a database connection using JDBC.
7. Explain the role of the DriverManager class in JDBC with suitable methods and examples.
8. Explain the Connection interface in JDBC. Discuss its important methods with examples.
9. Explain the Statement interface in JDBC and describe its commonly used methods.
10. Write and explain a Java program to establish a JDBC connection with a database.
11. Write a Java program to connect to a database and insert a student record using JDBC.
12. Write a Java program to retrieve and display records from a database using JDBC.
13. Write a Java program to update an existing student record using JDBC.
14. Write a Java program to delete a student record from a database using JDBC.
15. Explain ResultSet in JDBC. Discuss its purpose, cursor movement, and commonly used methods.
16. Differentiate between Statement and PreparedStatement with suitable examples.
17. Explain PreparedStatement and discuss its advantages over Statement.
18. Write and explain a JDBC program to perform insert, update, delete, and display operations on a Student table.
19. Explain the process of executing SQL queries using Statement and ResultSet in JDBC.
20. Explain SQLException in JDBC. Discuss common causes of database connection and query errors.
21. Explain the steps required to configure and use a JDBC driver in a Java application.
22. Explain DatabaseMetaData and ResultSetMetaData in JDBC with suitable examples.
23. Explain transaction management in JDBC. Discuss commit(), rollback(), and setAutoCommit().
24. Write a JDBC program to search for a student record using roll number and display the result.

Part B – Servlet Architecture and Life Cycle

- 25.** Explain the Servlet architecture in detail. Describe the role of the Servlet Container in managing Servlet objects and handling client requests.
- 26.** Explain the complete Servlet Life Cycle with a suitable diagram. Discuss the purpose and execution sequence of `init()`, `service()`, and `destroy()` methods.
- 27.** Differentiate between the Servlet Life Cycle and Thread Life Cycle. Explain how a Servlet Container handles multiple requests for the same Servlet.
- 28.** Explain the role of `ServletConfig` and `ServletContext`. Compare them with suitable examples.
- 29.** Explain the request-response processing model of a Servlet. Describe what happens from the time a client sends an HTTP request until the response is returned.
- 30.** Explain the difference between `doGet()` and `doPost()` methods. Discuss their advantages, limitations, security implications, and appropriate use cases.
- 31.** Explain the role of the `HttpServletRequest` and `HttpServletResponse` objects in Servlet programming. Discuss commonly used methods of both interfaces.
- 32.** Explain how a Servlet Container manages Servlet instances. Why is normally only one Servlet instance created for a particular Servlet?
- 33.** Explain what happens during the first request and subsequent requests to a Servlet. Clearly distinguish between `init()` and `service()` execution.
- 34.** What happens if the `init()` method of a Servlet throws an exception? Explain how the Servlet Container handles this situation.
- 35.** Explain why database connections and other expensive resources should be initialized and released carefully during the Servlet Life Cycle, with reference to `init()` and `destroy()`.
- 36.** Explain the relationship among the Servlet Container, `ServletConfig`, `ServletContext`, Request, Response, and Session during request processing.
- 37.** A Servlet receives requests from 100 users simultaneously. Explain how the Servlet Container can handle these requests without creating 100 Servlet objects.
- 38.** What is a Cookie? Explain the complete process of creating, sending, receiving, and deleting a Cookie in Servlet programming.

Part C – Cookies and Session Tracking

- 39.** Explain the limitations of Cookies as a session-tracking mechanism. Why might an application require an alternative mechanism?
- 40.** Differentiate between Session Cookies and Persistent Cookies. Explain the role of a cookie's maximum age.
- 41.** Explain how Cookies are transmitted between the client and server using HTTP request and response headers.

42. Suppose a user disables Cookies in the browser. Discuss the effect on a Servlet application that depends entirely on Cookies for session tracking.
43. Explain the security issues associated with storing sensitive information in Cookies. What precautions should be taken?
44. What is Session Tracking? Why is it required in HTTP-based web applications?
45. Explain and compare the major Session Tracking Techniques supported in Servlet-based applications: Cookies, URL Rewriting, Hidden Form Fields, and HttpSession.
46. Compare Cookies and HttpSession with respect to storage location, security, data capacity, lifetime, and server-side management.
47. Explain how HttpSession maintains the identity of a client across multiple HTTP requests.
48. Explain the complete life cycle of an HttpSession, including session creation, attribute management, timeout, and invalidation.
49. Differentiate between `setAttribute()`, `getAttribute()`, `removeAttribute()`, and `invalidate()`. Explain their use in session management.
50. A user logs into a web application successfully. After navigating to another Servlet, the application shows the user as not logged in. Identify possible causes related to session tracking and explain how you would troubleshoot the problem.
51. An application stores the username using a Cookie. The user closes the browser and opens the application again. Explain under what conditions the username may still be available.
52. A web application has disabled Cookies because of browser restrictions. Explain how you can maintain user sessions using URL Rewriting.
53. Consider an online shopping application. Explain how Servlet session management can be used to maintain a user's shopping cart across multiple requests.
54. A student logs in successfully but can directly access the Welcome Servlet without authentication after copying its URL. Explain the problem and describe how session-based authentication can prevent unauthorized access.

Part D – RequestDispatcher and Servlet Navigation

55. Explain the purpose of RequestDispatcher in Servlet programming. Differentiate between `forward()` and `include()`.
56. Differentiate between `RequestDispatcher.forward()` and `response.sendRedirect()` with respect to request object, response object, URL in browser, number of requests, processing location, performance, and data sharing.
57. A Servlet processes login information and needs to display a result using another Servlet. Explain when you would use `forward()` and when you would use `sendRedirect()`.
58. Why does the browser URL generally remain unchanged when `RequestDispatcher.forward()` is used but changes when `sendRedirect()` is used?
59. Explain how request attributes can be transferred from one Servlet to another using RequestDispatcher.

Part E – Advanced / Analytical Servlet Concepts

- 60.** “HTTP is stateless, whereas web applications require state.” Explain this statement and discuss how Servlet technology solves this problem.
- 61.** Explain why Servlets are considered reusable server-side components. Discuss the role of the Servlet Container in their management.
- 62.** Compare Cookies, URL Rewriting, Hidden Form Fields, and HttpSession as session-tracking mechanisms. Which mechanism would you prefer for a secure login application and why?
- 63.** Explain the possible problems of using instance variables in a Servlet when multiple users access the Servlet.
- 64.** A Servlet application uses a single Servlet instance to serve multiple clients. Explain the multithreading implications of this architecture and discuss how shared data should be handled safely.